

Half Steps		1hs	2hs	3hs	4hs	5hs	6hs	7hs	8hs	9hs	10hs	11hs	12hs
Degree	1	b2	2	b3	3	4	#4/b5	5	b6	6	b7	7	1
Interval	Tonic	minor Second	Major Second	minor Third	Major Third	Perfect Fourth	Tritone\Aug Fourth	Perfect Fifth	minor Sixth	Major Sixth	minor Seventh	Major Seventh	Octave

Scales		
Major	1 - 2 - 3 - 4 - 5 - 6 - 7	w - w - h - w - w - w - h
Major Pentatonic	1 - 2 - 3 - 5 - 6	w - w - w+h - w - w+h
Melodic Minor	1 - 2 - b3 - 4 - 5 - 6 - 7	w - h - w - w - w - w - h
Natural Minor (Aeolian)	1 - 2 - b3 - 4 - 5 - b6 - b7	w - h - w - w - h - w - w
Harmonic Minor	1 - 2 - b3 - 4 - 5 - b6 - 7	w - h - w - w - h - w+h - h
Minor Pentatonic	1 - b3 - 4 - 5 - b7	w+h - w - w - w+h - w
Japanese Scales		
Yo	1 - 2 - 4 - 5 - 6	Quality : Major
Akebono	1 - 2 - b3 - 5 - b6	Quality : Minor
Iwato	1 - b2 - 4 - b5 - b7	Quality : Minor / half diminished 7

Tonic	= I, iii, vi	~ Stable.
Sub-Dominant	= ii, IV	~ Unstable, often preceding dominant, or chords that move away from the tonic and typically transition to tension chords.
Dominant	= V(V7), vii dim	~ Unstable, pointing towards tonics.

Modes - KEY C		
1	C Ionian	C-D-E-F-G-A-B-C
2	D Dorian	D-E-F-G-A-B-C-D
3	E Phrygian	E-F-G-A-B-C-D-E
4	F Lydian	F-G-A-B-C-D-E-F
5	G Mixolydian	G-A-B-C-D-E-F-G
6	A Aeolian	A-B-C-D-E-F-G-A
7	B Locrian	B-C-D-E-F-G-A-B

Common Progressions	
ii - V7 - I	
iii - VI7 - ii - V7 - I	
I - VII - VI - V7	
I - VI7 - ii7 - V7	
I - vi - IV - V	
I - V - vi - IV	
around the circle of fifths	

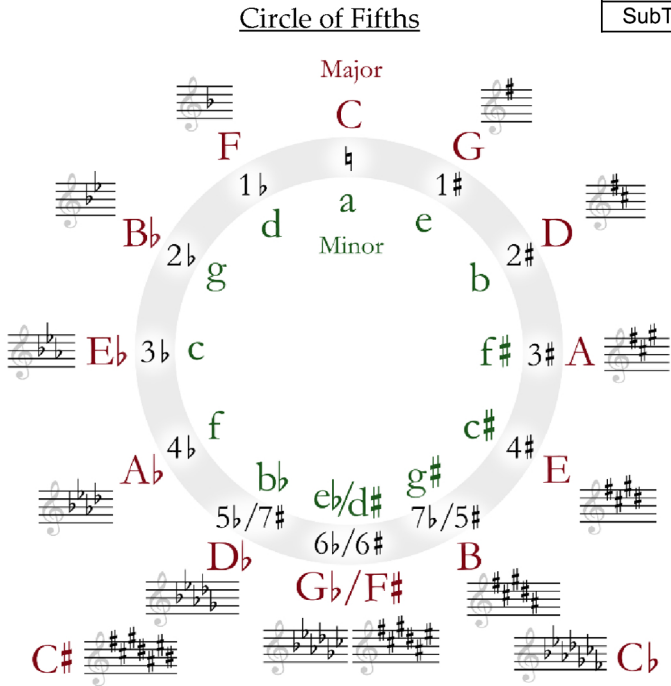
Parallel modes	
(modes/scales that share the same root note)	
Major Modes - C Major - CDEFGABC	
C Ionian	C-D-E-F-G-A-B-C
C Lydian	C-D-E-F#-G-A-B-C (#4)
C Mixolydian	C-D-E-F-G-A-Bb-C (b7)
Minor Modes - A minor - ABCDEFGA	
A Dorian	A-B-C-D-E-F#-G-A (#6)
A Phrygian	A-Bb-C-D-E-F-G (b2)
A Aeolian	A-B-C-D-E-F-G-A
A Locrian	A-Bb-C-D-Eb-F-G-A (b2 and b5)

Modal interchange - Taking chords from the Parallel mode	
C Ionian Chords	- Cmaj, Dmin, Emin, Fmaj, G7, Amin, Bdim
C Lydian Chords	- Gmaj, Amin, Bmin, Cmaj, D7, Emin, F#dim
C KEY progression with borrowed chord from lydian - Cmaj, Dmin, D7, Gmaj, Bmin, Cmaj (D7, Bmin, Gmaj are from lydian).	

Japanese Progressions	
IV - V7 - iii - vi or IV - iii - vi	(Merry christmas Mr lawrence - Sakamoto) (One Summer's Day - Hisaishi).
IV - V7 - vi	(Merry christmas Mr lawrence - Sakamoto) (Joe Hisaishi).
IV - V7 - VI	~Ending on VI (Picardy third - raising the third a semitone on an ending cadence - resulting in our major 6 ending chord).
bVII	(often after descending diatonic cadence or ending).

Chord substitutions	
1) Replace V with bVII.	
2) Secondary Dominant : V chord of V chords. (Key C : V chord is G7 - V chord of G is D7) (D7 is secondary dominant).	
3) Tritone Sub : Root of the V chord is X, replace the chord with a 7th chord whose root is a tritone away from X. (Works on dominant chords) G7 becomes Db7.	
4) Median Sub : Replace a chord whose root note is X with a chord whose root note is a third above or a third below X.	
5) ii - V Sub : Substitution for common ii V I progression.	
5a) 'Backdoor' : iv7 - bVII7 - I.	
5b) 'Tritone Sub' : vi7 - tritone7 from dominant - I.	

Degree	I	ii	iii	IV	V	vi	VII
Function	Tonic	SuperTonic	Mediant	Subdominant	Dominant	SubMediant	LeadingTone
						Natural Min -	bVII
							SubTonic



Major	
Major	1 3 5
Major6	1 3 5 6
Major 6/9	1 3 5 6 9
Major7	1 3 5 7
Major9	1 3 5 7 9
Major11	1 3 5 7 9 11
Major13	1 3 5 7 9 11 13

Minor	
Minor	1 b3 5
Minor6	1 b3 5 6
Minor7	1 b3 5 b7
Minor7b5/ hdim	1 b3 b5 b7
Minor9	1 b3 5 b7 9
Minor11	1 b3 5 b7 9 11
Minor13	1 b3 5 b7 13

Dominant	
Dominant7	1 3 5 b7
Dominant9	1 3 5 b7 9
Dominant11	1 3 5 b7 9 11
Dominant13	1 3 5 b7 9 11 13

Add	
Add2/Add9	1 3 5 2
Add4/Add11	1 3 5 4

Sus	
Sus2	1 2 5
Sus4	1 4 5

Extentions/Alterations	
Dominant7#9	1 3 5 b7 #9
Dominant7b9	1 3 5 b7 b9
Dominant7#5	1 3 5 b7 #5
Dominant7b5	1 3 5 b7 b5

Diminished	
Diminished	1 b3 b5
Diminished7	1 b3 b5 bb7

Augmented	
Augmented	1 3 #5
Augmented7	1 3 #5 b7